# Algorithms assignment 2

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**Problem** (2 Problem 5)**.** Given $7$ jobs with corresponding processing times,

| Job $j$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Processing Time $p_j$ | 4 | 7 | 2 | 9 | 12 | 5 | 3 |

The jobs will be processed on $3$ identical machines without interruption. All jobs are avliable at time $0$. (Model on slide 13, Chapter 2)

a) Use the local search algorithm where we have all jobs on machine $1$ in the beginning (in the same order as they are given). When you move a job, move it to the machine it can start earliest. I there is a tie, use the machine with the lowest number. What is the result, both time when the last job finish and the actual schedule?
b) Use the Greedy algorithm on the same data. When have all jobs finished?
c) Use the Longest processing time rule. When have all jobs finished?
d) Give a lower bound on the best solution and give a short explanation?
e) Is it possible to achieve the answer in d) (Explain)?

This is the classical identical parallel-machine makespan minimization problem, denoted

$$P_3 \| \|C_{\max}.$$

We want to find a schedule minimizing

$$C_{\max} = \max_{i \in [3]} L_i,$$

where

$$L_i := \sum_{j \text{ assigned to machine } i} p_j$$

is the total processing time on machine $i$.

There are two obvious immediate lower bounds on $C^*_{\max}$, the optimal makespan.

First, no machine can finish before the longest job assigned to it is done,

$$C^*_{\max} \geq \max_j p_j,$$

and the machines must collectively finish all the jobs so

$$C^*_{\max} \geq \frac{1}{3} \sum_{j=1}^{7} p_j.$$

*Solution to (a).* The initial schedule gives all seven jobs to machine $1$ in the described order by

$$M_1 = \underbrace{1}_{4} \quad \underbrace{1}_{4} \quad \underbrace{2}_{7} \quad \underbrace{3}_{2} \quad \underbrace{4}_{9} \quad \underbrace{5}_{12} \quad \underbrace{6}_{5} \quad \underbrace{7}_{3}.$$

Note: This accidentally looks a lot like fractions, but I am just illustrating the processing times $p_j$.

We get initial loads

$$(L_1, L_2, L_3) = (42, 0, 0)$$

with $C_{\max} = 42$.

We can identify that it is advantegous, or at least not deleterious, to swap a job with processing time $c$ from $M_i$ to $M_\iota$ if

$$\max\left(\{L_k \mid k \neq i, \iota\} \cup \{L_i - c, L_\iota + c\}\right)$$

is less than or equal to

$$\max_i L_i.$$

If $M_i$ is the unique machine for which $L_i = C_{\max}$ we find that it is advantageous to swap if

$$L_j - c < C_{\max}.$$

So we begin by looking at $j = 1$. $M_2$ ■